VOLUME 2A — THEORY NOTES: DESIGN APTITUDE, DESIGN FUNDAMENTALS & VISUAL REASONING

Each topic follows the same 12-point format:
Concept · Simple explanation · Key facts · Rules/principles · Common traps · How NID tests it · Example · Practice Qs · Answers · Explanations · Difficulty · Priority

Practice-question answers are given right below each set, so you can self-check immediately.

PART 1: DESIGN FUNDAMENTALS

1.1 Elements of Design

Concept: the elements are the basic "ingredients" every visual is built from. There are eight: **point, line, shape, form, colour, texture, space, value**.

Simple explanation & key facts

Element	What it is	What it does	Design example
Point	The smallest mark; a position	Attracts the eye; creates focus; many points make a pattern or tone (stippling)	A bindi on a blank wall; a map pin
Line	A point in motion; has direction	Guides the eye; divides; outlines; expresses mood. Horizontal = calm; vertical = strength/height; diagonal = action; curved = softness; zig-zag = tension	Road markings; the Nike swoosh
Shape	An enclosed 2D area	Geometric shapes (circle, square, triangle) feel man-made and stable; organic shapes feel natural and friendly	Traffic signs: triangle = warning, circle = mandatory/prohibitory, rectangle = information (Indian road-sign convention)
Form	3D shape with volume (sphere, cube, cone, cylinder)	Shown on paper through shading and perspective	A pressure cooker as cylinder + dome
Colour	Light reflected; hue, saturation, value	Mood, meaning, hierarchy, identification	Red for stop or danger
Texture	Surface quality, actual (felt) or visual (implied)	Realism, material identity, interest	Jute bag vs silk saree
Space	The area around and between objects. Positive space = the object; negative space = the background	Breathing room; figure–ground; depth	The FedEx arrow hidden in negative space
Value	Lightness or darkness	Creates form, depth and contrast	A pencil shading scale from 1 to 9

Rules / principles

- A line's *weight* communicates. Thick lines come forward or carry importance; thin lines recede or carry detail.

- Shapes carry meaning. Circles suggest unity or continuity; triangles suggest direction or danger; squares suggest stability or trust.
- Negative space is not empty; it is a design tool.

Common traps

- Confusing **shape (2D)** with **form (3D)**.
- Thinking texture must be real. **Visual texture** (drawn wood grain) counts.
- Thinking value only applies to black and white. Every colour has a value.

How NID tests it: MCQs ("Which element dominates this image?"); drawing tasks that use only one element ("Create a composition using only lines to express chaos"); negative-space logos.

Example: "Express *anger* using only lines." A good answer uses jagged zig-zags, sharp angles, heavy weight, dense clustering and diagonal direction.

Practice

1. Which element creates a sense of depth through gradation from light to dark? (a) Line (b) Value (c) Shape (d) Point
2. A logo where a hidden arrow sits between two letters relies mainly on: (a) texture (b) negative space (c) colour (d) form
3. Horizontal lines generally convey: (a) excitement (b) calm/rest (c) danger (d) growth
4. Drawing wood grain on a flat paper box is an example of: (a) actual texture (b) visual texture (c) form (d) value

Answers: 1-b, 2-b, 3-b, 4-b.
Explanations: (1) gradation of value models form and depth. (2) The FedEx arrow is formed by the space between E and x. (3) A horizontal line echoes the horizon and a body lying down: rest. (4) It looks textured but is smooth to touch, so it is visual texture.
Difficulty: Easy · **Priority:** HIGH

1.2 Principles of Design

Concept: principles are the "recipes": the ways elements are arranged to communicate.

Principle	Meaning	How to create it	Indian / everyday example
Balance	Visual weight evenly distributed. *Symmetrical* (formal), *asymmetrical* (informal but balanced), *radial* (from a centre)	Large light object vs small dark object; position from the centre	Taj Mahal (symmetrical); rangoli (radial); a modern poster (asymmetrical)
Contrast	Difference that creates interest and readability	Light/dark, big/small, warm/cool, thick/thin, smooth/rough	White text on a dark-green highway sign
Emphasis	One focal point dominates	Contrast, isolation, size, colour pop, converging lines	A red "SALE" on a grey poster
Hierarchy	Order of importance; what the viewer reads 1st, 2nd, 3rd	Size, weight, colour, position, spacing	A newspaper: headline → subheading → body
Proportion	Size relationship of parts to the whole	Golden ratio (~1:1.618), human proportions (adult ≈ 7.5–8 heads tall)	Correctly sized door vs person

Principle	Meaning	How to create it	Indian / everyday example
Scale	Size relative to a reference or to the human body	Exaggerate for drama; keep true scale for realism	The Statue of Unity looks huge next to people
Rhythm	Organised repetition that creates visual tempo: regular, alternating, progressive, flowing	Repeat elements with intervals	Pillars of a temple corridor; saree borders
Repetition	Using the same element again	Consistency, pattern, branding	Block-print motifs
Movement	The path the eye takes	Lines, diagonals, direction of gaze, gradients	A comic panel order; a figure's pointing hand
Unity	All parts feel like one whole	Common colour palette, style, alignment, proximity	A brand identity system
Variety	Differences that add interest	Change size, colour or shape within a theme	Different fonts used thoughtfully
Harmony	Parts are compatible and pleasing together	Related colours, similar shapes	Analogous earth tones in a Warli wall

Rules

- **Unity without variety is boring; variety without unity is chaos.**
- There should be **one** dominant focal point per composition. Two equal focal points compete.
- **Hierarchy** = emphasis applied to several levels.

Common traps: confusing **proportion** (part-to-part, within the object) with **scale** (relative to another object or the human); confusing **rhythm** (repetition *with tempo*) with plain repetition.

How NID tests it: "Identify the principle in the image" MCQs; "redesign this poster to improve hierarchy"; compositions with constraints ("show rhythm using only circles").

Practice

1. A small red dot placed off-centre on a large white page balancing a large grey square on the other side shows: (a) symmetrical balance (b) asymmetrical balance (c) radial balance (d) no balance
2. Size → weight → colour used to order headline, sub-head and body is: (a) unity (b) rhythm (c) hierarchy (d) harmony
3. A rangoli built around a central point shows: (a) radial balance (b) asymmetry (c) proportion (d) texture
4. A poster with seven fonts, ten colours and no alignment lacks: (a) variety (b) unity (c) contrast (d) movement
5. Gradually increasing circle sizes across a page creates: (a) regular rhythm (b) progressive rhythm (c) radial balance (d) unity

Answers: 1-b, 2-c, 3-a, 4-b, 5-b.
Explanations: (1) Different objects are balanced by visual weight: a saturated small dot vs a large neutral mass. (2) Ordering by importance is hierarchy. (3) Elements radiate from a centre. (4) Too much variety destroys unity. (5) Progressive rhythm changes in steps.
Difficulty: Easy–Medium · **Priority:** HIGH

1.3 Colour Theory

Concept: a systematic way to choose and combine colours.

Key facts

Colour wheel (RYB, the painter's/art-school wheel used in NID-style questions)

```
                YELLOW (P)
        Y-Orange(T)       Y-Green(T)
   ORANGE(S)                   GREEN(S)
        R-Orange(T)       B-Green(T)
                RED(P)  ...  BLUE(P)
        R-Violet(T)       B-Violet(T)
                VIOLET(S)
P = primary   S = secondary   T = tertiary
```

- **Primary (RYB):** red, yellow, blue. **Secondary:** orange (R+Y), green (Y+B), violet (B+R). **Tertiary:** primary + adjacent secondary, e.g. red-orange, yellow-green, blue-violet.
- **Light (additive, screens): RGB.** R+G = yellow; G+B = cyan; R+B = magenta; R+G+B = white.
- **Print (subtractive): CMYK.** Cyan, magenta, yellow, key (black). C+M+Y in theory gives black (in practice a muddy brown, hence the K).
- **Hue** = the colour family name. **Saturation / chroma** = intensity or purity. **Value** = lightness or darkness.
- **Tint** = hue + white (pastel). **Shade** = hue + black. **Tone** = hue + grey.

Colour schemes

Scheme	Definition	Effect	Example
Monochromatic	One hue in different tints, tones and shades	Calm, unified	All blues
Analogous	2–5 neighbours on the wheel	Harmonious, natural	Yellow, yellow-orange, orange
Complementary	Opposite on the wheel: R–G, B–O, Y–V	Maximum contrast, vibrant	Indian cricket kit: blue + orange accents
Split-complementary	One hue + the two neighbours of its complement	Contrast with less tension	Blue + red-orange + yellow-orange
Triadic	3 hues equally spaced (120°)	Lively, balanced	R, Y, B or O, G, V
Tetradic / square	4 hues (2 complementary pairs)	Rich; hard to balance	R, G, O, B
Warm	Reds, oranges, yellows	Energy, appetite, advance visually	Food brands
Cool	Blues, greens, violets	Calm, recede visually	Hospitals, banks

Colour psychology (culture-dependent; Indian context noted)

- **Red:** energy, danger, love. In India it is auspicious, bridal and sindoor.
- **Saffron / orange:** spirituality, courage; it appears on the national flag.
- **Yellow:** happiness, caution, haldi, spring (Basant).
- **Green:** nature, safety, go, Islam, prosperity.
- **Blue:** trust, calm, technology; associated with Krishna.
- **White:** purity, peace. In India it is also the colour of mourning.
- **Black:** power, elegance, mourning (West), protection from the evil eye (*nazar* in India).
- **Purple:** luxury, royalty, creativity.

Rules

- **60-30-10 rule:** 60% dominant colour, 30% secondary, 10% accent.
- Warm colours advance and cool colours recede. Use this for depth in landscapes (**atmospheric perspective**: distant hills look bluer and paler).

- For readability, **value contrast** matters more than hue contrast. Red text on green has the same value and fails, especially for colour-blind viewers.
- About 8% of men have red–green colour blindness. Never use colour as the only signal; add a shape or label.

Common traps

- Mixing up the RYB wheel and the RGB light model. Red + green paint makes brown; red + green light makes yellow.
- "Tone" in everyday speech means many things, but in colour theory it means **hue + grey**.
- Complementary pairs depend on the wheel; NID-style questions assume RYB (red–green, blue–orange, yellow–violet).

How NID tests it: MCQs on complements and schemes; "colour this scene using only analogous colours"; "show the same street in a warm palette (morning festival) and a cool palette (rainy night)".

Example: "Design a poster for a hospital blood-donation drive." Use red as the 10% accent (the drop), lots of white for cleanliness (60%), and calm blue-grey for text (30%).

Practice

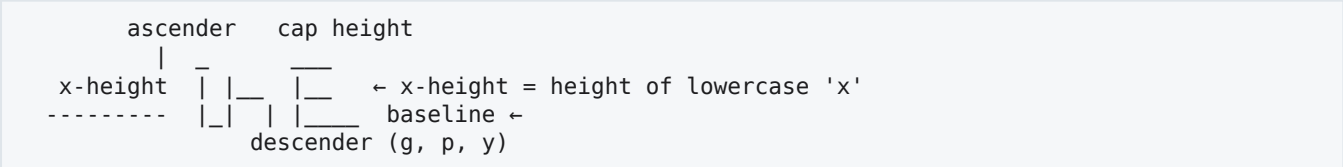
1. The complement of blue on the RYB wheel is: (a) yellow (b) orange (c) green (d) violet
2. Red + white produces: (a) shade (b) tint (c) tone (d) hue
3. Mixing red, green and blue light in equal intensity produces: (a) black (b) brown (c) white (d) grey
4. Yellow-green is: (a) primary (b) secondary (c) tertiary (d) complementary
5. A palette of light blue, mid blue and navy is: (a) analogous (b) monochromatic (c) triadic (d) complementary
6. In a landscape, distant mountains are painted bluish and paler because of: (a) linear perspective (b) atmospheric perspective (c) foreshortening (d) chiaroscuro

Answers: 1-b, 2-b, 3-c, 4-c, 5-b, 6-b.
Explanations: (1) Blue is opposite orange. (2) Adding white gives a tint (pink). (3) Additive mixing gives white. (4) A primary plus an adjacent secondary is tertiary. (5) One hue with value changes is monochromatic. (6) Air scatters light, so far-off objects lose contrast and turn bluish.
Difficulty: Easy · **Priority:** HIGH

1.4 Typography

Concept: the art of arranging type to make language readable, legible and expressive.

Key facts & terminology



Term	Meaning
Serif	Small strokes at the ends of letters (Times New Roman, Garamond, Georgia). Traditional, bookish; good for long print text.
Sans serif	No serifs (Helvetica, Arial, Futura, Gill Sans). Modern, clean; good on screens and signage.
Slab serif	Thick block serifs (Rockwell, Courier). Sturdy; typewriter feel.

Term	Meaning
Script	Handwriting-like (Brush Script). Elegant or personal.
Display / decorative	For headlines only; low legibility at small sizes.
Monospace	Every character the same width (Courier). Used for code.
Typeface vs font	A typeface is the design family (Helvetica). A font is one specific weight/style/size (Helvetica Bold 12 pt).
Kerning	Space between a *specific pair* of letters (AV, To)
Tracking	Uniform spacing across a *whole word or line*
Leading (say "ledding")	Vertical space between lines (baseline to baseline); from strips of lead in metal type
Alignment	Left (ragged right), right, centred, justified
Legibility	How easily individual *characters* can be told apart (a property of the typeface design)
Readability	How easily *text blocks* can be read (a property of the layout: size, leading, line length, contrast)
Hierarchy	Using size, weight and colour to order the information
Widow / orphan	A lonely word or line at the end or start of a paragraph or column
Point size	1 pt ≈ 1/72 inch

Rules

- Ideal body-text line length is about **45–75 characters**.
- Use a maximum of **2 typefaces** per layout (one display + one text).
- Leading of about **120–145%** of the type size suits body text.
- Use all caps for short labels only; long all-caps text is hard to read.
- Signage: sans serif, high contrast, large x-height. Highway signs use mixed case because word shapes help recognition at speed.

Common traps: kerning vs tracking; legibility vs readability; "font" vs "typeface".

How NID tests it: "Which is more legible?" MCQs; "write the word FRAGILE so it looks fragile" (expressive typography); signage design; lettering in posters.

Example: expressive typography. **MELT**: letters drip downward. **HEAVY**: thick, bottom-weighted, sinking. **ECHO**: repeated, fading letters.

Practice

1. Adjusting the space between "A" and "V" only is: (a) tracking (b) kerning (c) leading (d) alignment
2. Which typeface classification is generally preferred for wayfinding signage? (a) Script (b) Decorative (c) Sans serif (d) Blackletter
3. Leading refers to: (a) letter width (b) space between lines (c) space between words (d) stroke thickness
4. Helvetica Bold Italic 14 pt is best described as a: (a) typeface (b) font (c) type family (d) glyph set

Answers: 1-b, 2-c, 3-b, 4-b.
Explanations: (1) Kerning adjusts one pair. (2) Sans serif stays clear at a distance and at speed. (3) Leading is the line spacing. (4) A specific style and size is a font.
Difficulty: Easy–Medium · **Priority:** MEDIUM (HIGH for poster and signage tasks)

1.5 Composition

Concept: the arrangement of elements inside a frame.

Key techniques

- **Rule of thirds:** divide the frame 3×3 and place subjects on the lines or intersections ("power points"). This feels more dynamic than dead centre.



- **Symmetry:** formal, stable, ceremonial (temple fronts, logos).
- **Asymmetry:** dynamic and modern; balance it with visual weight.
- **Focal point:** where the eye lands first.
- **Negative space:** empty space around the subject that gives it importance.
- **Grid:** an invisible structure for alignment (columns, modules).
- **Leading lines:** roads, rails or rivers that pull the eye to the subject.
- **Framing:** doors, windows or arches that frame the subject.
- **Foreground–middle ground–background:** layering creates depth.
- **Odd numbers:** groups of 3 or 5 feel more natural than 2 or 4.
- **Golden ratio / spiral:** 1:1.618.
- **Visual weight:** large, dark, saturated, textured, isolated or high-placed elements feel heavier.

Rules for NID drawing boxes

1. Decide the viewpoint and horizon *before* drawing.
2. Put the main action on a thirds intersection.
3. Use overlapping to show depth.
4. Crop boldly. Letting objects run off the edge makes the scene feel larger.
5. Leave some breathing space; don't fill every millimetre with equal detail.

Common traps: centring everything; tangents (edges that just touch, creating confusion); a horizon line that cuts the drawing exactly in half; all objects the same size.

How NID tests it: every scene and poster is judged on composition; MCQs ask "which layout follows the rule of thirds?"

Practice

1. Placing the horizon on the lower third line emphasises: (a) the ground (b) the sky (c) the people (d) nothing
2. A railway track converging toward a distant station is an example of: (a) framing (b) leading lines (c) radial balance (d) texture
3. Why are tangents avoided? (a) They waste ink (b) They create visual confusion about depth (c) They are hard to draw (d) They add too much colour

Answers: 1-b, 2-b, 3-b.
Explanations: (1) Two-thirds of the frame becomes sky. (2) The converging rails guide the eye. (3) When edges touch exactly, it is unclear which object is in front.

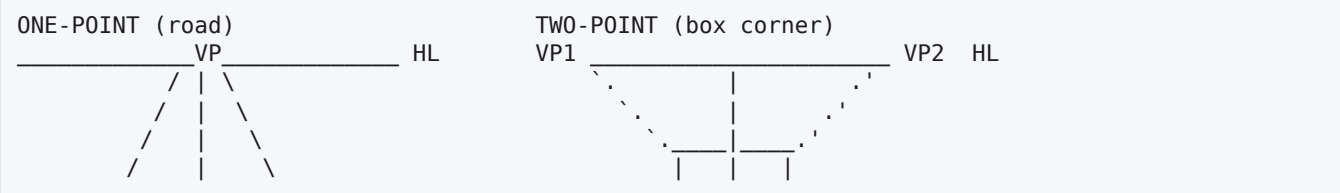
Difficulty: Easy · **Priority:** VERY HIGH (applied in every drawing)

1.6 Perspective

Concept: a way to draw 3D space on a 2D surface so that it looks real.

Key facts

- **Horizon line (HL)** = the viewer's eye level. Everything above the HL is seen from below; everything below it is seen from above.
- **Vanishing point (VP)** = the point on the HL where parallel lines appear to meet.
- **One-point perspective:** one VP. Used for frontal views of roads, corridors, railway tracks and rooms. Lines are horizontal, vertical, or converge to the VP.
- **Two-point perspective:** two VPs on the HL. Used for a building corner or a box seen edge-on. Verticals stay vertical.
- **Three-point perspective:** two VPs on the HL plus a third above (worm's-eye) or below (bird's-eye). Used for tall buildings or dramatic views.
- **Foreshortening:** objects pointing toward the viewer look compressed (an outstretched arm).
- **Atmospheric perspective:** distance reduces contrast and detail and shifts colour toward blue.
- **Size diminution:** identical objects get smaller with distance. **Overlap:** nearer objects hide farther ones.



Rules for people in perspective (very important for NID scenes)

- If the viewer is standing on flat ground, **the eye level (HL) passes through the heads of all standing adults of similar height, at any distance.** Draw farther people smaller, but keep their heads on the HL.
- If the HL is at waist height (viewer sitting), the HL passes through the waists of standing adults.
- Doors are about 2.1 m tall; an adult is about 1.6–1.75 m. Use doors and vehicles as scale references.

Common traps: several horizon lines in one scene; verticals that tilt in 1- or 2-point perspective; VPs placed too close together (heavy distortion); people floating above the ground plane.

How NID tests it: "Draw a street, platform or market from a given viewpoint"; "draw the view from a balcony or the top of a Ferris wheel" (bird's-eye); "the view of an ant" (worm's-eye); MCQs ("where is the VP?").

Practice

1. In a standing-eye-level street scene, where do the heads of distant standing adults fall? (a) Above the HL (b) On the HL (c) Below the HL (d) At the VP only
2. A tall skyscraper drawn looking straight up needs: (a) 1-point (b) 2-point (c) 3-point (d) isometric
3. Which perspective keeps all vertical lines vertical and uses two VPs? (a) one-point (b) two-point (c) three-point (d) aerial
4. Isometric drawing differs from perspective because: (a) parallel lines stay parallel (b) it uses three VPs (c) objects shrink with distance (d) it has no 3D effect

Answers: 1-b, 2-c, 3-b, 4-a.
Explanations: (1) The eye-height rule. (2) A third VP is needed for the vertical convergence. (3) The two-point definition. (4) Isometric is a parallel projection; nothing converges.
Difficulty: Medium · **Priority:** VERY HIGH

1.7 Gestalt Principles of Visual Perception

Concept: the mind organises parts into wholes ("the whole is other than the sum of its parts"). These principles come from early-20th-century German psychologists (Wertheimer, Köhler, Koffka).

Principle	Meaning	Example
Proximity	Things near each other are seen as a group	Grouped buttons on a remote
Similarity	Things that look alike (colour, shape) form a group	A team in the same jersey colour
Closure	The mind completes incomplete shapes	The WWF panda, IBM stripes
Continuity	The eye follows smooth paths	A curving road
Figure–ground	We separate an object from its background	Rubin's vase (faces or vase)
Common fate	Elements moving together are grouped	A flock of birds
Symmetry	Symmetrical elements are seen as one unit	Brackets [ ]
Prägnanz (simplicity)	We read the simplest interpretation	The Olympic rings read as 5 circles, not complex shapes

Trap: closure vs continuity. Closure fills **gaps in a shape**; continuity follows **lines or paths**.
How NID tests it: identify the principle used in a logo; design a logo using closure.

Practice: 1. The WWF panda logo uses: (a) proximity (b) closure (c) common fate (d) continuity. 2. Rubin's vase demonstrates: (a) similarity (b) figure–ground (c) symmetry (d) rhythm.
Answers: 1-b (the outline is incomplete and the mind fills it in); 2-b (it switches between two readings).
Difficulty: Easy · **Priority:** MEDIUM

1.8 Visual Communication: Pictograms, Signage, Posters, Logos, Packaging

Pictogram rules

1. Use a simple silhouette, ideally recognisable in under 1 second.
2. Keep one idea per sign.
3. Use consistent line weight, corner radius and stance across a set (think Otl Aicher's 1972 Munich Olympic pictograms).
4. Use a standard grid (e.g. a square with consistent margins).
5. Don't rely on text. A pictogram should work for non-readers and in many languages.
6. Use colour conventions: red = prohibition/fire, yellow = warning, green = safety/exit, blue = mandatory/information.
7. Prohibition = a red circle with a diagonal slash from top-left to bottom-right, over a black symbol.

Signage for narrow Indian streets (directly relevant: DAT 2026 asked for 5 signboards)

- **Placement:** projecting (blade) signs perpendicular to the wall, so they can be seen down a narrow lane; at a height clear of heads and vehicles (about 2.4 m+).

- **Visibility:** high contrast, big icons, short words; bilingual (local language + English) or icon-led; reflective or lit at night.
- **Content:**
  - one-way lane
  - two-wheelers only / no four-wheelers
  - dead end ("no through road")
  - pedestrian priority
  - shop direction
  - public toilet / water tap
  - temple / masjid ahead, "silence zone"
  - no parking
  - honking prohibited
  - low overhead wires
  - emergency exit route
  - house-number directory
- **Materials:** weather-resistant (powder-coated metal, enamel); compact.

Poster anatomy: headline (big, 3–6 words) → visual (one strong image) → supporting text → call to action → logo / date / venue. Use the Z-pattern or F-pattern of eye movement.

Logo types: wordmark (Google, Coca-Cola), lettermark (IBM, HP, LIC), pictorial (Apple, Twitter bird), abstract (Nike swoosh, Pepsi globe), mascot (Amul girl, Air India Maharaja), emblem (Harley-Davidson, Starbucks), combination mark (Burger King).
Good logo qualities: simple, memorable, timeless, versatile (works in black and white, small sizes, embroidered, on a phone icon), appropriate.

Packaging functions: protect, preserve, contain, communicate, convenience (open, pour, reseal, carry), sustainability (mono-material, compostable, reusable).

How NID tests it: signage sets, posters (2025: a crow's-viewpoint juice poster), symbols, logos, packaging redesign.
Difficulty: Medium · **Priority:** VERY HIGH

1.9 Human-Centred Design (HCD) & Design Thinking

Concept: solving problems by first understanding the people affected.

The 5-stage design thinking process (Stanford d.school model)

Stage	What you do	NID-paper equivalent	Example: grandmother's travel kit (DAT 2026 theme)
1. Empathise	Observe, interview, feel the user's situation	Imagine the user's day; list pains	Grandma: weak eyesight, knee pain, medicines at fixed times, needs a shawl, a fear of losing tickets
2. Define	Write a clear problem statement: "[User] needs [need] because [insight]"	Underline the real problem	"Grandma needs to find medicines and essentials quickly without bending or searching, because she tires easily and has poor eyesight."
3. Ideate	Generate many ideas without judging; then select	Quick thumbnails of 3–5 ideas	A colour-coded pouch organiser; a pill box with alarm; a trolley bag with a seat; a neck pouch for tickets
4. Prototype	Make it tangible: sketch, mock-up	The final labelled drawing	Drawing of the organiser with labels: "large-print tags", "magnetic flap", "reflective strip"

Stage	What you do	NID-paper equivalent	Example: grandmother's travel kit (DAT 2026 theme)
5. Test	Get feedback; iterate	Add a "how it helps" note or a scenario frame	Frame: grandma opening the pouch on the train easily

Other models: the Double Diamond (UK Design Council, 2005): **Discover → Define → Develop → Deliver**. It alternates divergent and convergent thinking. IDEO's HCD: Inspiration → Ideation → Implementation.

Key concepts

- **Persona:** a fictional but realistic user profile.
- **User journey map:** the steps the user goes through, with emotions at each step.
- **Pain points:** moments of frustration.
- **Affordance (Don Norman):** a property that shows how to use an object. A handle affords pulling; a flat plate affords pushing.
- **Signifier:** a perceivable cue (a "PUSH" label, an arrow).
- **Feedback:** the system's response (a click sound, a light).
- **Mapping:** the relationship between controls and effects (stove knobs laid out like the burners).
- **Constraints:** limits that prevent errors (a USB-C plug fits either way; an asymmetric plug fits one way only).
- **Poka-yoke (mistake-proofing):** from Japanese manufacturing.
- **Universal / inclusive design:** usable by the widest range of people (ramps, OXO Good Grips peelers, kerb cuts).
- **Accessibility:** design for disabilities (tactile paving: lines = direction, dots = warning/stop).

Don Norman's book: *The Design of Everyday Things* (originally *The Psychology of Everyday Things*, 1988). It popularised "Norman doors", doors whose design confuses people about whether to push or pull.

Dieter Rams' 10 principles of good design: good design

1. is innovative
2. makes a product useful
3. is aesthetic
4. makes a product understandable
5. is unobtrusive
6. is honest
7. is long-lasting
8. is thorough down to the last detail
9. is environmentally friendly
10. is as little design as possible ("Weniger, aber besser" = "less, but better")

Common traps

- Jumping to the solution without a problem statement.
- Designing for yourself instead of the user.
- Over-tech solutions (an app for everything) where a simple physical solution works better.
- Ignoring cost and context in India: electricity cuts, water scarcity, crowding, multiple languages, literacy levels.

How NID tests it: "Identify problems faced by X and propose a solution"; "redesign Y for elderly users"; the neem-branch neighbour problem (2025). Examiners look for a visible **problem → insight → solution** chain.

Practice

1. A flat metal plate on a door signals "push". This is an example of: (a) feedback (b) affordance (c) mapping (d) constraint
2. The Double Diamond's four stages are: (a) Plan, Do, Check, Act (b) Discover, Define, Develop, Deliver (c) Empathise, Define, Ideate, Test (d) Research, Sketch, Build, Sell
3. Stove knobs arranged in the same layout as the burners show good: (a) feedback (b) mapping (c) affordance (d) branding
4. Tactile paving with raised dots usually means: (a) go straight (b) warning / stop (c) turn left (d) a shop entrance

Answers: 1-b, 2-b, 3-b, 4-b.
Explanations: (1) The plate's form suggests the action, so it affords pushing (a signifier in Norman's later terminology; NID-level questions accept "affordance"). (2) The Design Council model. (3) The spatial correspondence is mapping. (4) Dots mark hazards or crossings; bars give direction.
Difficulty: Easy–Medium · **Priority:** VERY HIGH

1.10 Creative Thinking Techniques

Technique	How it works	Quick NID example
Brainstorming (Alex Osborn)	Quantity over quality; no judgement	20 uses of a brick in 3 minutes
SCAMPER (Bob Eberle)	**S**ubstitute, **C**ombine, **A**dapt, **M**odify/Magnify, **P**ut to other use, **E**liminate, **R**everse/Rearrange	Umbrella: *Combine* with a torch; *Reverse* it so it folds inside-out to keep water in
Lateral thinking (Edward de Bono)	Escape obvious patterns; provocation ("Po"), random entry	"What if trains had no seats?" leads to standing supports with lean pads
Six Thinking Hats (de Bono)	White = facts, Red = feelings, Black = caution, Yellow = benefits, Green = creativity, Blue = process	Evaluate a solution from 6 angles
Mind mapping (Tony Buzan)	Radial association	Theme "monsoon" → umbrellas, potholes, chai, frogs, floods, greenery
Forced connection	Combine two unrelated things	Kettle + bird = a whistling kettle shaped like a bird
Changing scale / viewpoint	See the problem as an ant, a giant, a crow	The 2025 crow's-viewpoint poster
Biomimicry	Learn from nature	Velcro (burrs); the Shinkansen nose (kingfisher beak); termite-mound cooling (Eastgate Centre, Harare, by Mick Pearce)
Reverse brainstorming	"How could we make this worse?" then flip the answers	"How to make a bus stop terrible?" → no shade, no info, so add both

Torrance's creativity dimensions: fluency (number of ideas), flexibility (variety of categories), originality (rarity), elaboration (detail). NID's "multiple uses" and "make as many as you can" tasks test fluency and flexibility.

Trap: listing 10 ideas from one category (e.g. ten kinds of container) shows fluency but low flexibility. Spread across categories: tool, toy, art, shelter, music, game.

Practice: List the SCAMPER letter for each. (i) Using a bicycle wheel as a pottery wheel. (ii) A pen with a built-in stylus. (iii) Removing the cord from headphones.
Answers: (i) P (put to other use); (ii) C (combine); (iii) E (eliminate).

Difficulty: Easy · **Priority:** HIGH

1.11 Ergonomics, Usability & Sustainability (quick reference)

Ergonomics

- Anthropometry is the measurement of human bodies. Design usually covers the 5th–95th percentile of users.
- Seat height ≈ knee height (about 40–45 cm for adults). Table ≈ 72–75 cm. Kitchen counter ≈ 85–90 cm (Indian standards often ~80–85 cm). Door ≈ 2.1 m high × 0.9 m wide. Stair riser ≈ 15–18 cm, tread ≈ 25–30 cm.
- Grip: a handle diameter of 30–40 mm suits a power grip.
- Wheelchair ramp slope: about 1:12 is the widely used maximum.

Usability (Jakob Nielsen's heuristics, simplified)

- visibility of system status
- match with the real world
- user control
- consistency
- error prevention
- recognition over recall
- flexibility
- minimalist design
- error recovery
- help

Sustainability

- The R's: Refuse, Reduce, Reuse, Repair, Refurbish, Recycle, Rot (compost).
- Life-cycle thinking: raw material → manufacture → transport → use → end of life.
- **Cradle to Cradle** (McDonough & Braungart, 2002): products as nutrients, with no waste.
- **Circular economy** vs linear "take-make-dispose".
- **Design for disassembly**: screws, not glue; mono-materials.
- **Indian traditions:**
  - kulhads (clay cups)
  - leaf plates (pattal, sal or areca leaves)
  - steel tiffins
  - jute and cotton bags
  - stepwells and jaali for passive cooling
  - kantha quilting from old saris (upcycling)
- **Greenwashing:** a false or exaggerated environmental claim.

Practice

1. An adult chair seat height is closest to: (a) 25 cm (b) 43 cm (c) 65 cm (d) 80 cm
2. Kantha embroidery traditionally reuses: (a) plastic (b) old saris and cloth (c) leaves (d) metal
3. Designing a product with screws instead of glue supports: (a) obsolescence (b) design for disassembly (c) greenwashing (d) ergonomics only

Answers: 1-b, 2-b, 3-b.
Explanations: (1) Seat height roughly matches the lower-leg length. (2) Kantha layers old cloth with running stitch. (3) Screws allow repair and material separation.
Difficulty: Easy · **Priority:** MEDIUM–HIGH

PART 2: VISUAL REASONING

Each type below has: rules, a worked example, traps and practice. All practice items are text/ASCII-solvable. More practice is in Vol 4B.

2.1 Mirror Images

Rule: a mirror reverses the direction **perpendicular to the mirror**.

- **Vertical mirror** (on the right or left side of the object): left ↔ right swap; top and bottom stay the same.
- **Horizontal mirror** (below or above, as in a "water image"): top ↔ bottom swap; left and right stay the same.

Letters symmetrical about a vertical axis (unchanged in a vertical mirror): **A, H, I, M, O, T, U, V, W, X, Y**.
Letters symmetrical about a horizontal axis (unchanged in a water image): **B, C, D, E, H, I, K, O, X**.
Letters unchanged in both: H, I, O, X.

Worked example: the vertical-mirror image of **"NID"**. Reverse the order to D-I-N, then flip each letter left–right: D becomes a backward D, I stays I, N becomes a backward N. Result: **ᗡIИ** (written right-to-left). Students often forget to flip the individual letters.

Clock mirror trick: mirror time = **11:60 − actual time** (for a vertical mirror). 3:15 → 11:60 − 3:15 = **8:45**. If the actual time is between 11:00 and 12:59, use 23:60 − time instead. Water-image (horizontal mirror) time has no single-subtraction rule that works for all times, so sketch the clock and flip it top-to-bottom. Only the vertical-mirror formula is worth memorising.

Traps: forgetting to reverse the letter order; mixing up which axis flips; numbers 2, 3, 4, 5, 6, 7, 9 are asymmetric. 0, 1 (simple stroke) and 8 are symmetric about a vertical axis.

Practice

1. Which word looks the same in a vertical mirror? (a) MOM (b) DAD (c) NUN (d) BOB
2. A clock shows 2:20. Its mirror image shows: (a) 9:40 (b) 10:40 (c) 9:20 (d) 10:20
3. Which letter looks the same in a water image? (a) A (b) M (c) E (d) U

Answers: 1-a, 2-a, 3-c.
Explanations: (1) M and O are vertically symmetric, and the palindrome order keeps MOM unchanged; D and B are not vertically symmetric; N is not. (2) 11:60 − 2:20 = 9:40. (3) E is symmetric about a horizontal axis.
Difficulty: Easy–Medium · **Priority:** MEDIUM

2.2 Rotation

Rule: rotation turns a figure without flipping it. 90° clockwise: top → right → bottom → left → top. A rotated figure can **never** become its mirror image. "L" rotated is always some L-shape and can never become a "⅃".

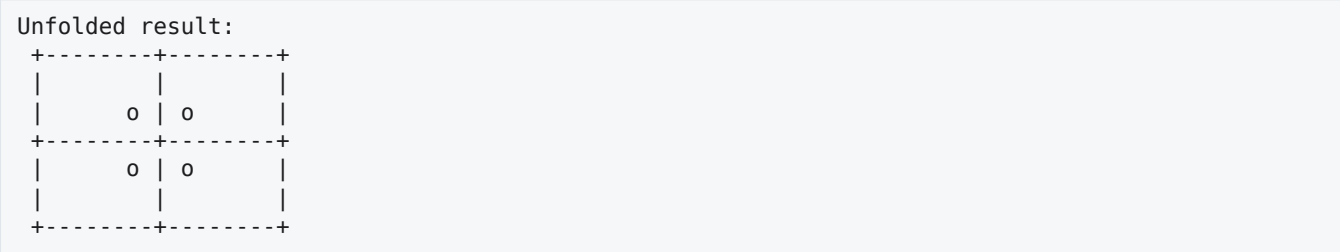
Worked example: the letter "F" rotated 180° looks like an upside-down, backward F, but it is still not a mirror F. To test an option, mentally rotate it back.

Trap: choosing a mirror image when the question asks for a rotation.
Practice: An arrow points North. It is rotated 135° clockwise, then 45° anticlockwise. Where does it point?
Answer: net 90° clockwise → **East**.
Difficulty: Easy · **Priority:** MEDIUM

2.3 Paper Folding & Cutting

Rule: each fold doubles the layers. A hole punched through *n* layers gives *n* holes when unfolded, placed symmetrically about each fold line. **Unfold in reverse order.**

Worked example: a square sheet is folded in half (left over right), then folded in half again (top over bottom). One hole is punched near the folded corner. Unfolded, there are **4 holes**, arranged symmetrically about both the vertical and horizontal centre lines.



(If the hole is near the corner where both folds meet, the four holes cluster near the centre, as shown.)

Traps: a hole punched *on* a fold line creates fewer separate holes (it becomes one elongated or merged hole); diagonal folds reflect across the diagonal; forgetting the reverse order.

Practice

1. A paper is folded in half 3 times. One hole is punched through all layers. How many holes appear when unfolded?
2. A square is folded once along its diagonal and a semicircle is cut on the folded edge (the cut sits on the fold). Unfolded, what is the cut shape?

Answers: 1 → **8** (2^3 layers). 2 → a **full circle** centred on the diagonal (the semicircle mirrors across the fold).
Difficulty: Medium · **Priority:** HIGH (if objective items return)

2.4 Cubes, Dice & Nets

Key facts

- A cube has 6 faces, 12 edges and 8 vertices.
- On a standard die, opposite faces sum to 7 (1–6, 2–5, 3–4).

- **Net rule:** in a straight row of faces in a net, faces **one apart** (with one face between them) are **opposite** in the folded cube. There are **11 distinct nets** of a cube.

```
Net example (cross):
      [1]
  [2] [3] [4] [5]
      [6]
Opposites: 3–5 (one apart in the row), 2–4 (one apart), 1–6 (above/below the same square 3)
```

Painted-cube formulas (a big cube of edge n cut into n^3 small cubes, all outer faces painted):

- 3 faces painted = **8** (the corners)
- 2 faces painted = **12(n − 2)** (edges)
- 1 face painted = **6(n − 2)²** (face centres)
- 0 faces painted = **(n − 2)³** (inside)

Check for n = 3: 8 + 12 + 6 + 1 = 27 ✓.

Two-position dice rule: if two views of the same die share **one** common visible face and that face is in the same position, the other faces in the same positions are opposite. Otherwise, use rotation reasoning.

Practice

1. A 4×4×4 painted cube is cut into 64 small cubes. How many have exactly 2 faces painted?
2. In the cross net above, which face is opposite 2?
3. How many small cubes have no paint in a 5×5×5 painted cube?

Answers: 1 → 12(4 − 2) = **24**. 2 → **4**. 3 → (5 − 2)³ = **27**.
Difficulty: Medium · **Priority:** HIGH (if objective items return)

2.5 Counting Figures (triangles, squares)

Formulas

- Triangles from a single apex with n lines fanning down to a base: **n(n − 1)/2** (where n counts all lines from the apex, including the two outer sides).
- Squares in an n×n grid: **$1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$**. For a 4×4 grid: 30.
- Rectangles in an m×n grid: **C(m+1, 2) × C(n+1, 2)**. For a 3×3 grid: 6 × 6 = 36 (this includes squares).
- A square with both diagonals contains **8 triangles** (4 small + 4 made of two halves).

Practice

1. How many squares are in a 3×3 grid?
2. A triangle has 4 lines from its top vertex to the base (including the two sides). How many triangles?
3. How many rectangles (including squares) in a 2×3 grid?

Answers: 1 → 1 + 4 + 9 = **14**. 2 → 4×3/2 = **6**. 3 → C(3,2) × C(4,2) = 3 × 6 = **18**.
Difficulty: Medium · **Priority:** MEDIUM

2.6 Pattern & Series Completion

Common transformations:

- rotation by a fixed angle each step (45°, 90°)
- an element added or removed each step

- shading moving clockwise
- size increasing
- alternating pairs
- the number of sides increasing (triangle → square → pentagon)
- positions cycling through corners

Method: track **one** element at a time across the series. Write down its rule. Repeat for each element.

Practice: series of polygons: triangle (3), square (4), pentagon (5), hexagon (6), ? → **heptagon (7 sides)**. Dot inside moving: top-left, top-right, bottom-right, bottom-left, ? → **top-left** (it cycles clockwise).
Difficulty: Easy–Medium · **Priority:** MEDIUM

2.7 Embedded / Hidden Figures, Silhouettes, Shadows

- **Embedded figures:** find a simple shape hidden within a complex one. Look for its distinctive angles first.
- **Silhouettes:** the outline shape alone must identify the object. Good silhouette = a characteristic profile (a teapot's spout and handle, side-on).
- **Shadows:** a shadow falls *away* from the light. Its length depends on the light's height (low sun = long shadow at morning or evening; noon = short). All shadows in one scene point consistently from one light source. The cast shadow touches the object where the object touches the ground.
- **Orthographic views:** front, top and side (plan, elevation). From an isometric object, picture what each view shows. Hidden edges are drawn as dashed lines in technical drawing.

Practice

1. In the morning, the sun is in the East. Which way does your shadow fall? → **West**.
2. A cylinder lying on its side: its top view is a **rectangle**; its end view is a **circle**.
3. A cone standing on its base: the top view is a **circle with a centre point**; the front view is a **triangle**.
   Difficulty: Easy–Medium · **Priority:** MEDIUM–HIGH (also needed for drawing)

PART 3: DESIGN APTITUDE — OBSERVATION & SENSITIVITY

3.1 Observation

Concept: noticing details others miss, such as wear patterns, colours, proportions and how people actually use things.

NID-style observation facts worth knowing (common MCQ / drawing checks)

- An Indian ₹ symbol has **two horizontal lines** at the top (designed by D. Udaya Kumar, adopted in 2010).
- The national flag ratio is **2:3**. The Ashoka Chakra has **24 spokes** and is navy blue.
- A standard 4-legged chair: front legs are often splayed; most plastic chairs (monobloc) are stackable.
- Traffic light order, top to bottom: **red, amber, green**.
- A ceiling fan (Indian) usually has **3 blades**. Table fans have a guard.
- On a QWERTY keyboard the top-left letter row starts **Q W E R T Y**. The F and J keys have bumps for touch-typing.
- Analogue clocks in adverts often show **10:10** (it frames the logo and looks like a smile).

- Pencil hardness: H = hard and light; B = black and soft. HB is the middle.
- Shoelaces: the eyelets are usually in two columns with pairs at matching heights.
- A bicycle's chain is on the **right** side (drive side).
- Indian vehicles drive on the **left**, so the steering wheel is on the right.

Drill: every day, choose one object. Draw it from memory, then observe the real object and correct the drawing in red. Track your errors.

Priority: HIGH

3.2 Visual Sensitivity / Aesthetic Judgement

- Recognising good vs poor proportion, alignment and spacing; consistency of style; appropriateness of colour.
- **NID-style task:** "Which of these 4 layouts is best for a book cover and why?" Judge it by hierarchy, contrast, negative space and appropriateness.

Priority: MEDIUM

Quick self-test for Volume 2A (answers below)

1. Name the three primaries of light.
2. What is the difference between proportion and scale?
3. Which design thinking stage writes the problem statement?
4. In two-point perspective, how many vanishing points lie on the horizon line?
5. What does "affordance" mean?
6. How many distinct nets does a cube have?
7. How many faces of a 3×3×3 painted cube's centre small cube are painted?
8. Name the four stages of the Double Diamond.
9. Which letters look the same in both mirror and water images?
10. What is the 60-30-10 rule?

Answers

1. Red, Green, Blue.
2. Proportion compares parts within an object; scale compares the object's size to another reference (usually the human body).
3. Define.
4. Two.
5. A property of an object that suggests how it is used.
6. 11.
7. Zero.
8. Discover, Define, Develop, Deliver.
9. H, I, O, X.
10. 60% dominant, 30% secondary, 10% accent colour.

NEXT VOLUME: Volume 2B — Logical Reasoning, Quantitative Aptitude & Verbal Ability Notes (`02B-VOL2-THEORY-REASONING-QUANT-VERBAL.md`)